

Analog and Digital Communication (EE3003)

Final Exam

Date: December 24, 2025

Course Instructor(s)

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Total Time (Hrs): 3
Total Marks: 100
Total Questions: 5

Roll No

Section

Student Signature

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1. Attempt all the questions. Attempt all parts of the same question together.
2. Show all the steps with the help of diagrams and mathematical equations.

CLO # 02: Apply the concept of sampling, analog amplitude and frequency modulation and demodulation techniques.

Q1:

[20 marks]

- (a) Name three modulation schemes that are bandwidth efficient. [5]
- (b) Draw and mathematically expoit Vestigial Sideband (VSB) transmitter (modulator) and receiver (demodulator). Clearly label each block and describe its function. [7]
- (c) The transfer function of a vestigial sideband filter $H_i(\omega)$ is shown in Fig. 1 given below. The carrier frequency is $f_c = 10$ kHz and the baseband signal bandwidth is 4 kHz. Determine and sketch the corresponding transfer function of the equalizer filter $H_o(\omega)$ at the receiver. [8]

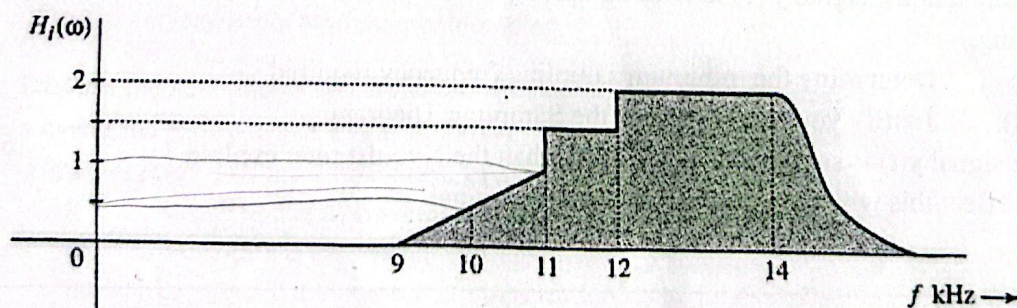


Figure 1. Vestigial Side Band Filter $H_i(\omega)$

CLO 02

Q2: Given that the frequency sensitivity $k_f = 200,000\pi$ and phase sensitivity $k_p = 10$. Assume the bandwidth of $m(t)$ to be equal to the third harmonic frequency of $m(t)$. **[20 marks]**

For the modulating signal

$$m(t) = \sin(2000\pi t)$$

(a) Estimate and calculate the bandwidth of frequency modulated signal B_{FM} and the phase modulated signal B_{PM} . **[8]**

(b) Analyze and comment on the effect on bandwidth for the following cases: **[12]**

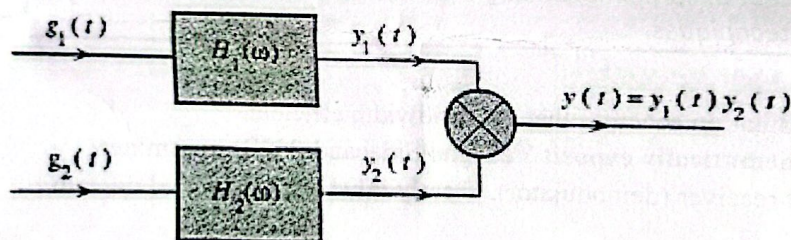
- (i) when the amplitude of $m(t)$ is doubled.
- (ii) when $m(t)$ is time expanded by a factor of 2.
- (iii) when the frequency of $m(t)$ is doubled.

CLO 02

Q3: The signals $g_1(t) = 10^4 \text{rect}(10^4 t)$ and $g_2(t) = \delta(t)$ are applied at the inputs of ideal low-pass filters with frequency responses, **[20 marks]**

$$H_1(\omega) = \text{rect}\left(\frac{\omega}{40,000\pi}\right), \quad H_2(\omega) = \text{rect}\left(\frac{\omega}{20,000\pi}\right)$$

The outputs are then multiplied to produce $y(t) = y_1(t) y_2(t)$.



CLO 02

Figure 2. Low Pass Filter

- (a) Determine the outputs $y_1(t)$ and $y_2(t)$ of the two filters. **[4]**
- (b) Find the Nyquist rate of $y_1(t)$, $y_2(t)$ and the combined signal $y(t)$. **[4]**
- (c) Assume that the signal $y(t)$ is to be sampled and reconstructed without aliasing. **[6]**
 - i) Determine the minimum sampling frequency required. **[2]**
 - ii) Justify your answer using the Sampling Theorem.
- (d) If the signal $y(t)$ is sampled at a rate lower than the Nyquist rate, explain what effect this will have on the reconstructed signal. **[6]**

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CLO # 03: Analyze different digital communication techniques, modulation schemes, and coding methods to evaluate system performance under noise and channel impairments.

Q4: CLO 03 [20 marks]

- (a) A band-limited video signal $v(t)$ of 4 MHz bandwidth is sampled at a rate of $33\frac{1}{3}\%$ higher than the Nyquist rate. The maximum allowable error in the sample amplitude (i.e. the maximum quantization error) is 0.45% of the peak amplitude v_p . Assuming binary encoding PCM, answer the following: [12]
- i) Find the Nyquist rate and the Actual rate of sampling.
 - ii) Find the number of bits, n to be transmitted per sample and the quantization levels, L .
 - iii) Define maximum information rate theorem and find the minimum transmission bandwidth of the channel to transmit 32 such encoded binary signals (also known as T1 line in the Bell Telephone system).
- (b) Draw the block diagram of Delta-Modulated (DM) transmitter (modulator) and receiver (demodulator) with mathematical exposition. [4]
- (c) A DM system is designed to operate at eight (8) times the Nyquist rate. The signal bandwidth B at its input port is 4 kHz and the quantized step Δ is 350 mV. Assuming $f_m = 2$ kHz message sinusoidal input, find the maximum amplitude A_m of this tone that avoids slop overload. [4]

$$\frac{0.45}{100} m_p = \frac{m_p}{L}$$

2 integ.
left

Q5: CLO 03 [20 marks]

- (a) Consider the binary sequence "0 1 1 0 0 1 0 1", find and draw the waveforms of the following signaling formats: [12]
- i) Binary Unipolar NRZ signaling
 - ii) Bipolar RZ (AMI) signaling
 - iii) Manchester (Split-Phase) signaling
 - iv) Differential Manchester encoding
- (b) Discuss the advantages and disadvantages of Bipolar NRZ scheme. [4]
- (c) Answer the following as **True** or **False**. For the **False** ones, give the correct answer. For the **True** ones, justify your answer. [4]
- i) Binary encoded waveforms occupy more bandwidth than $m(t)$.
 - ii) Baud-rate is always higher than bit rate.
 - iii) Increasing the number of quantization levels has no effect on bandwidth.
 - iv) Line codes are used to reduce bandwidth requirement.